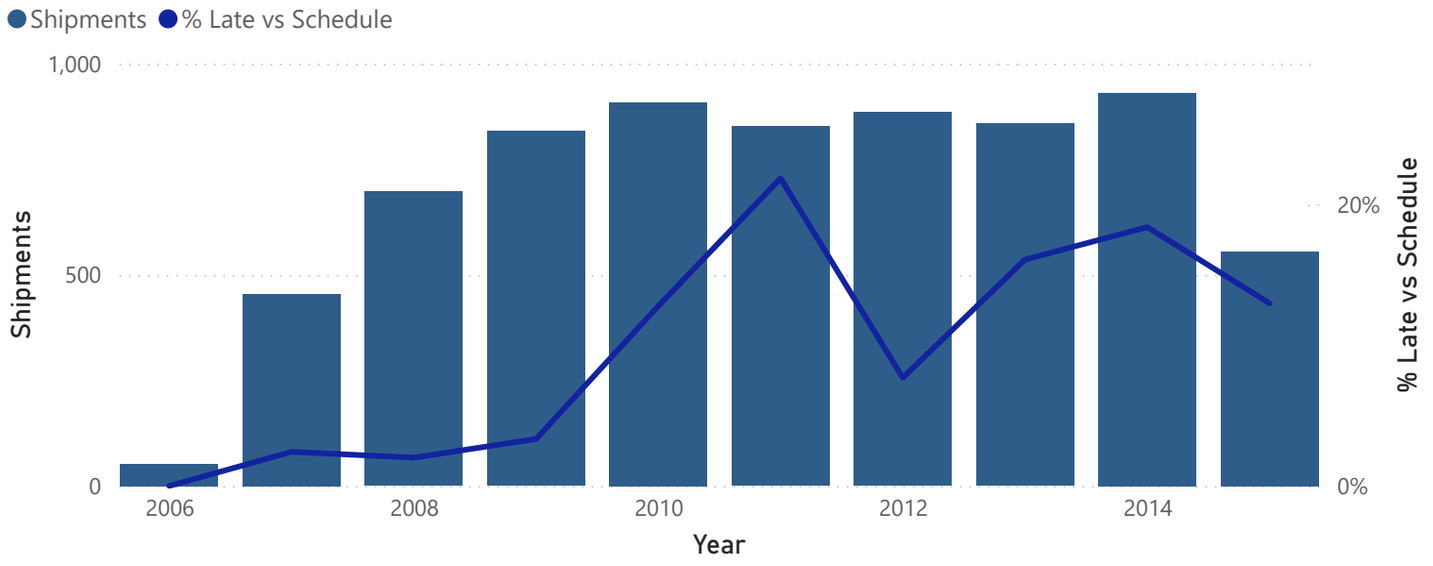
Global Health Supply Chain Performance

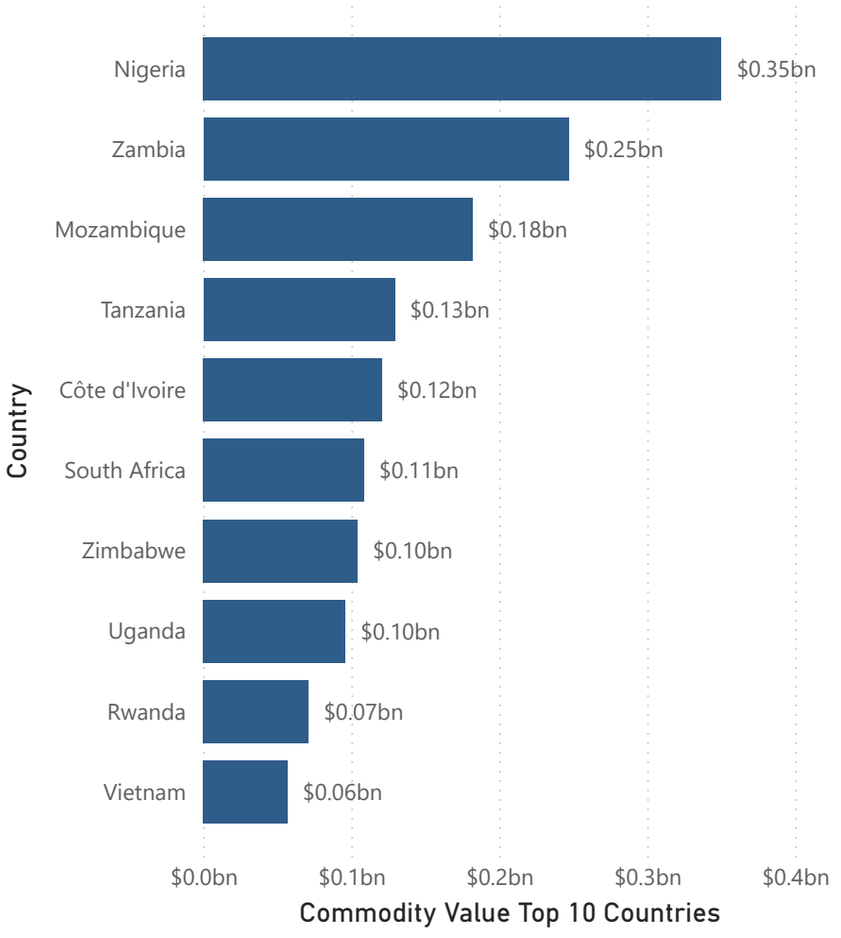
USAID / PEPFAR SCMS shipments, 2006-2015. Late = delivered after the scheduled date; per the data dictionary the scheduled date is not the client-promised date.

Commodity value	Shipments	% late vs schedule	Avg PO-to-delivery days (Direct Drop)	Freight cost (recorded)
$1.63bn	7,030	11.4%	113.0	$68.82M

Shipments and % late vs schedule by delivery year



Top 10 countries by commodity value



Delivery year

Year

All

Shipment mode

Shipment Mode

All

Fulfilment route

Fulfill Via

All

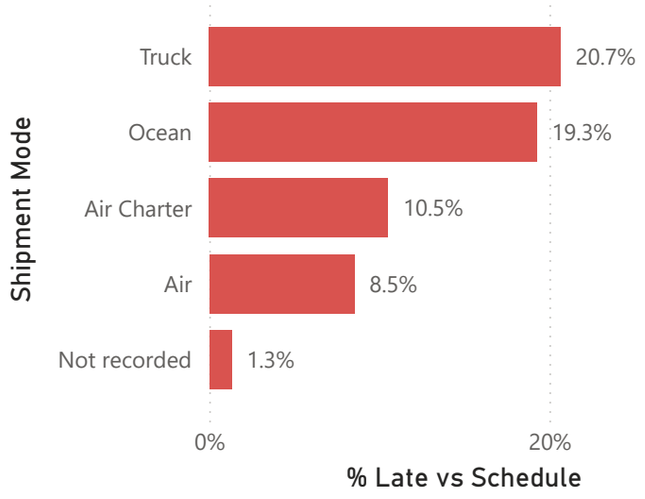
How to read this page

Volume and lateness are counted per shipment (7,030), value per line item (10,324). Freight and weight are recorded once per shipment, so they come from the shipment table.

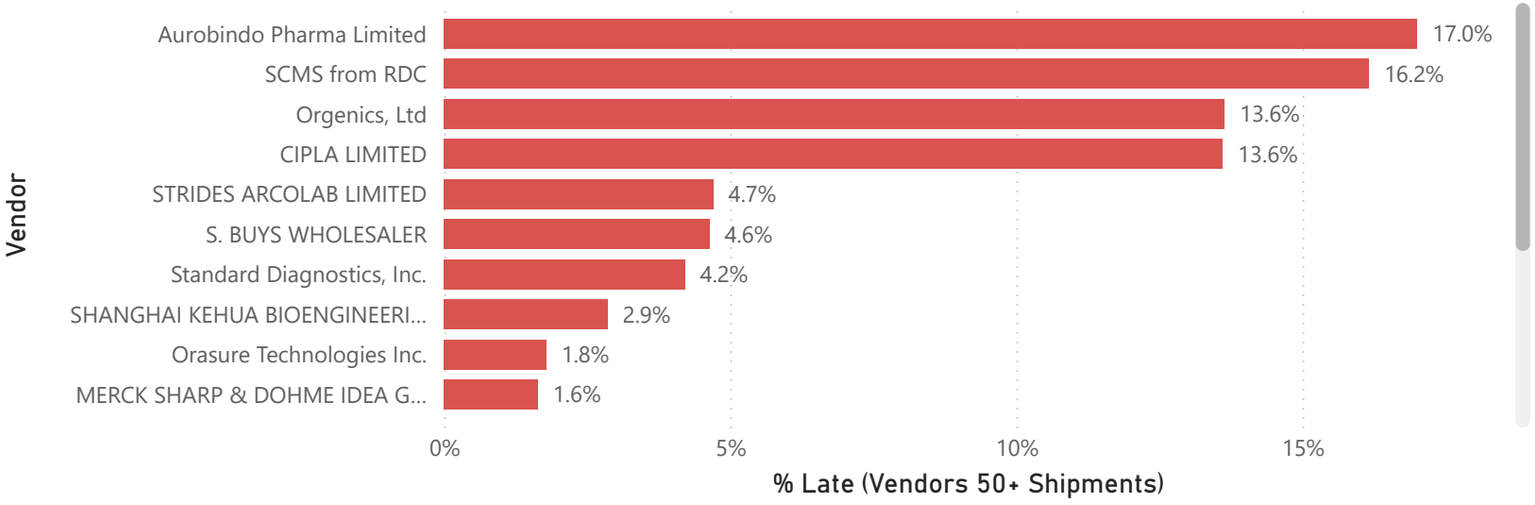
Delivery performance

Which modes, vendors, routes and countries deliver later than scheduled.

% late vs schedule by shipment mode



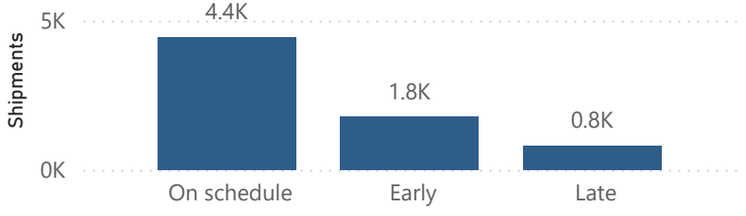
% late vs schedule by vendor (vendors with 50+ shipments)



Direct Drop vs From RDC

Route	Shipments	% late	Avg lead time (days)
Direct Drop	3,590	6.9%	113.0
From RDC	3,440	16.2%	
Total	**7,030**	**11.4%**	**113.0**

Shipments by delivery status



% late vs schedule by country and year (countries with 100+ shipments)

Country	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	**Total**
Congo, DRC							15.8%	11.5%	29.4%	33.3%	**21.2%**
Côte d'Ivoire	0.0%	1.0%	5.3%	8.5%	12.5%	20.3%	3.5%	13.3%	15.9%	33.3%	**10.8%**
Ethiopia	0.0%	11.1%	6.5%	0.0%	10.7%	15.0%	0.0%	0.0%	0.0%	0.0%	**6.1%**
Guyana	0.0%	12.5%	0.0%	0.0%	4.3%	0.0%	0.0%	21.7%	28.6%	0.0%	**8.5%**
Haiti	0.0%	5.9%	0.0%	0.0%	21.7%	30.6%	4.0%	2.2%	6.0%	1.7%	**6.7%**
Kenya			0.0%	11.8%	30.4%	5.6%	19.0%	0.0%	0.0%		**13.6%**
Mozambique		0.0%	2.9%	2.1%	17.9%	27.7%	18.6%	19.2%	28.0%	34.4%	**17.8%**
Nigeria	0.0%	7.3%	1.1%	2.2%	9.7%	24.4%	8.2%	29.4%	20.0%	8.9%	**12.1%**
Rwanda		3.6%	0.0%	0.0%	0.0%	0.0%	4.9%	4.5%	25.0%	15.9%	**8.0%**
South Africa		0.0%	5.4%	6.8%	31.9%	28.2%	40.9%	0.0%	0.0%	0.0%	**19.1%**
Tanzania	0.0%	0.0%	2.3%	0.0%	9.7%	13.5%	6.6%	11.5%	32.9%	11.1%	**11.5%**
Uganda	0.0%	0.0%	0.0%	2.5%	7.4%	39.0%	4.4%	22.3%	15.2%	6.7%	**11.8%**
Total	**0.0%**	**2.4%**	**2.0%**	**3.3%**	**12.8%**	**21.9%**	**7.7%**	**16.1%**	**18.4%**	**13.0%**	**11.4%**

Freight and cost

Freight per kg and freight as a share of commodity value, using only shipments where both numbers were recorded.

Freight cost (recorded)

$68.82M

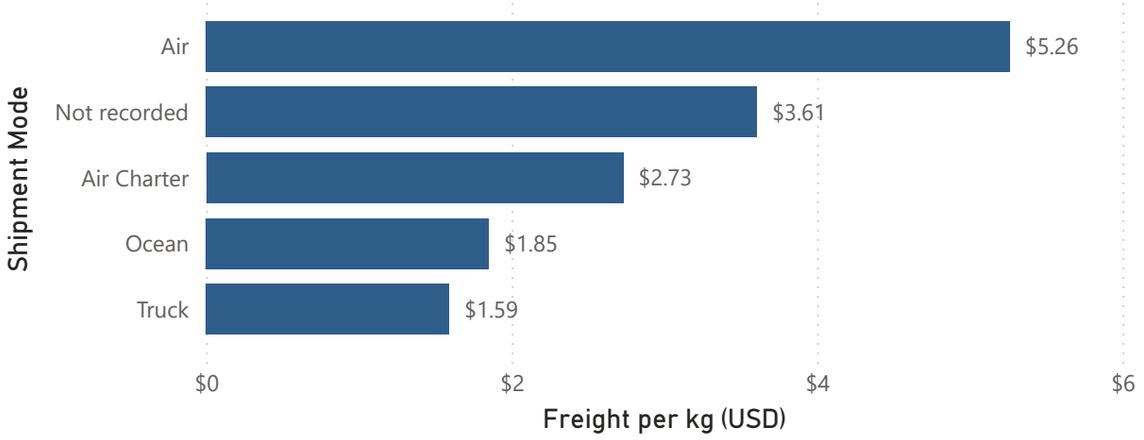
Freight per kg

$3.24

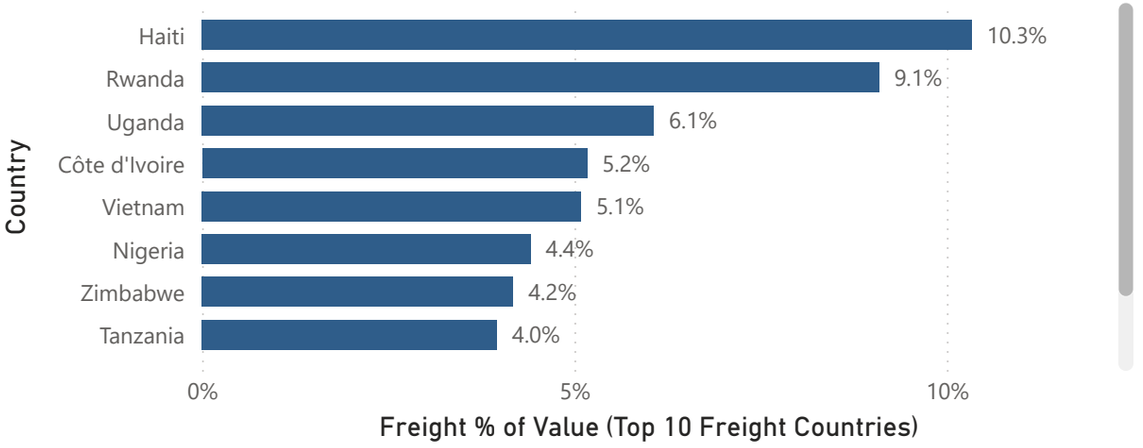
Freight as % of commodity value

4.5%

Freight cost per kg by shipment mode



Freight as % of value, top 10 countries by freight spend



Cost vs reliability by shipment mode

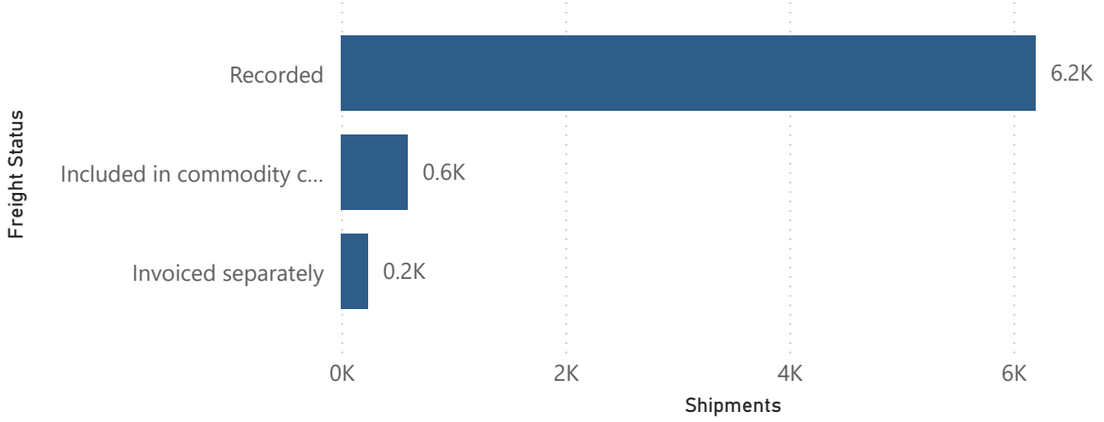
Shipment Mode	Shipments	Freight per kg (USD)	Freight % of Value	% Late vs Schedule	Avg PO to Delivery Days
Air	4,541	$5.26	7.3%	8.5%	110.4
Air Charter	494	$2.73	3.8%	10.5%	79.1
Not recorded	226	$3.61	3.5%	1.3%	2.3
Ocean	327	$1.85	3.1%	19.3%	181.3
Truck	1,442	$1.59	2.2%	20.7%	49.7
Total	**7,030**	**$3.24**	**4.5%**	**11.4%**	**113.0**

Data quality

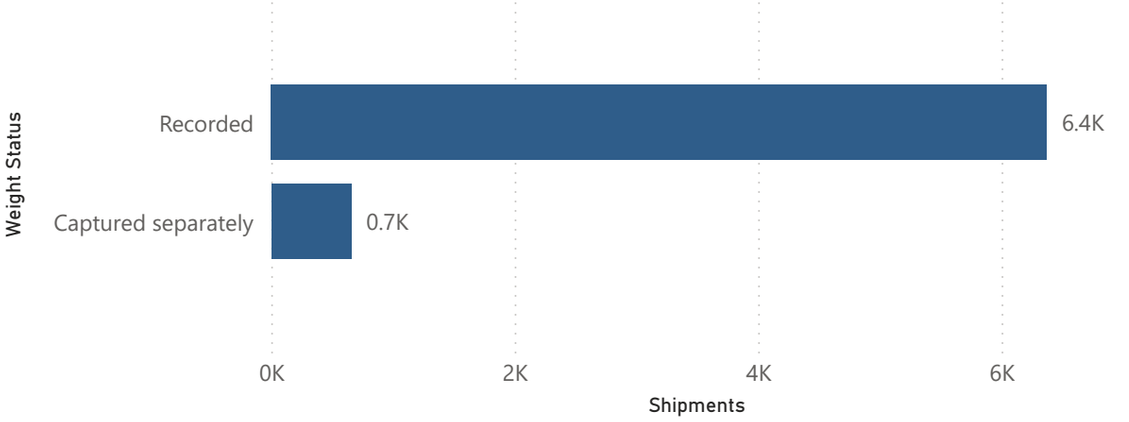
What is missing or unreliable in the source data, and how it was handled.

Shipments without separate freight	Shipments without weight	Line items	Shipments
11.8%	9.4%	10,324	7,030

Shipments by freight status



Shipments by weight status



Lead time by shipment mode: check 'Not recorded'

Shipment Mode	Shipments	Avg PO to Delivery Days	Median PO to Delivery Days
Air	4,541	110.4	95
Air Charter	494	79.1	64
Not recorded	226	2.3	17
Ocean	327	181.3	174
Truck	1,442	49.7	22
Total	**7,030**	**113.0**	**98**

Cleaning steps

1. Imported as UTF-8 so names like Cote d'Ivoire stay intact.
2. Parsed two date formats (28-Sep-06 and 9/11/14) with an en-US locale.
3. Kept placeholder text (Date Not Captured, Freight Included in Commodity Cost) in status columns, then set the values to blank.
4. Split line items (10,324) from shipments (7,030): weight and freight are stored once per shipment.
5. Flag: shipments with no recorded mode average about 2 days from PO to delivery, which is not plausible. Treat that group with caution.